

DAX Lab 03 – Complete Learning Guide

1. What is DAX?

- DAX (Data Analysis Expressions) is a formula language used in Power BI to perform calculations and data analysis.
- It works on tables and columns inside a data model.
- DAX is different from Excel because it works with relationships and context.

2. Calculated Column vs Measure

- Calculated Column: Row-by-row calculation, stored in model, uses Row Context.
- Measure: Aggregated result, calculated dynamically, uses Filter Context.
- Use columns for per-row logic like Gross Sales. Use measures for totals like Total Sales.

3. Row Context vs Filter Context

- Row Context: Exists in calculated columns. Each row is evaluated separately.
- Filter Context: Created by slicers, filters, visuals. Affects measures dynamically.
- CALCULATE modifies filter context.

4. Core Row-Level Calculations

- Gross Sales = Quantity × Unit Price
- Discount Amount = Gross Sales × Discount%
- Net Sales = Gross Sales – Discount
- Total Cost = Quantity × Unit Cost
- Profit = Net Sales – Total Cost
- Profit Margin = Profit / Net Sales

5. Aggregated Measures

- Total Net Sales = SUM(Net Sales)
- Total Profit = SUM(Profit)
- Total Quantity = SUM(Quantity)

- Unique Customers = `DISTINCTCOUNT(CustomerID)`
- Average Order Value = Total Net Sales / Total Orders

6. Important Functions

- `SUM()`, `COUNT()`, `DISTINCTCOUNT()`, `MIN()`, `MAX()`
- `IF()` and `SWITCH(TRUE())` for logical classification.
- `DIVIDE()` for safe division (avoids divide-by-zero).
- `CALCULATE()` to modify filter context.
- `ALL()` removes filters for percentage of total.
- `FILTER()` applies conditional filtering.

7. Percentage of Total Concept

- Region % of Total = Region Sales / Grand Total Sales
- Use `ALL()` in denominator to remove region filter.
- This ensures denominator remains constant.

8. Dashboard Design Guidance

- Top: KPI Cards (Total Sales, Profit, Margin).
- Middle: Sales by Region (Bar Chart).
- Category Analysis (Column Chart).
- Slicers: Region, Category, Gender.
- Use consistent colors and clean layout.

9. Common Mistakes

- Using calculated column instead of measure for totals.
- Forgetting `DIVIDE()` for safe division.
- Not understanding filter context before writing `CALCULATE`.